Elxie MVP: Home Edition

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Summary

The MVP is the Elxie Home edition, proven with a pilot of 10 robots. Elxie is a smart toy for the toy market: a small ESP32-S3 robot with a hand remote and a web app. Children first play five robot games in the app, and finishing all game levels unlocks Elxie Blocks, a Scratch-style block coding editor that programs the same robot.

The MVP must show three things on real hardware, in real homes:

1. Children can set up and play the games without adult help after the first connection.
2. Finishing the games unlocks Blocks, and children write and run their own programs on the robot.
3. All 10 robots and remotes run one firmware build, are paired, and work reliably for the whole pilot.

Who it's for

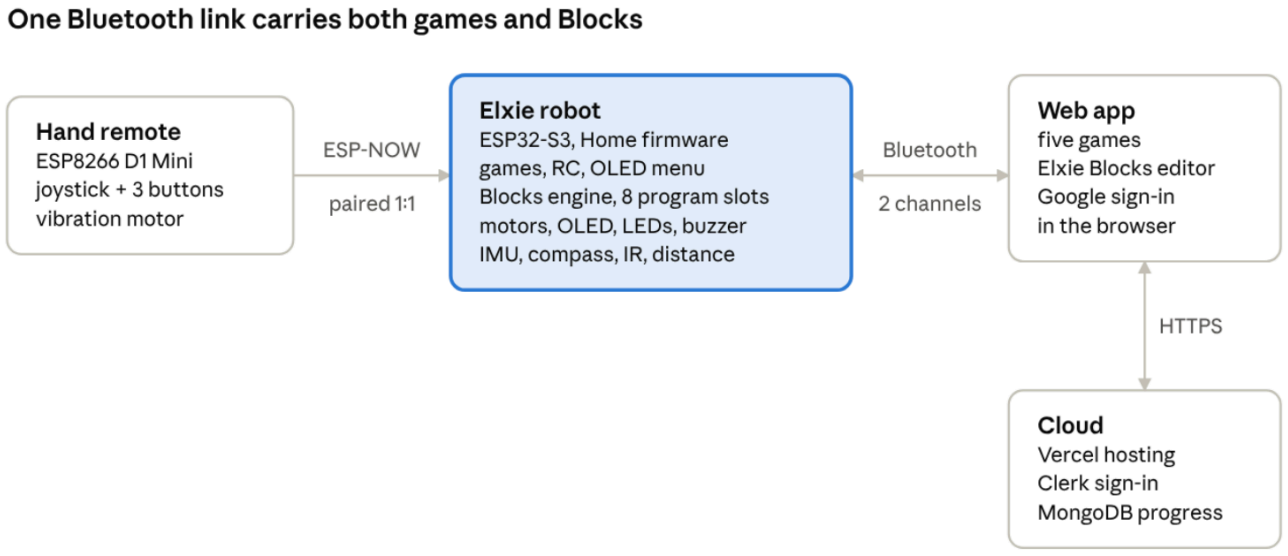
The primary user is a child aged 5 to 12 playing at home; the buyer and helper is a parent.

User	Needs from the MVP
Child, 5 to 8	Play with the remote and simple games; big buttons, short words, instant robot reactions
Child, 9 to 12	Clear game goals, then real programming with blocks: sensors, loops, events
Parent	Easy first setup, progress saved to the child's Google account, safe robot behaviour

Schools are a later edition (Blocks only, always unlocked) and are not part of this MVP.

The product

Elxie is three devices and one cloud service: the robot does all the work, the remote and the web app drive it.



Elxie system · robot, remote, web app, cloud

The remote talks only to its paired robot. The app uses one Bluetooth connection with two channels: one for games and RC, one for Blocks programs, so a child never reconnects to switch.

User journey

A child goes from unboxing to their first own program in eight steps; steps 1 to 3 happen once.

1. **Switch on.** The robot shows its welcome face; the remote buzzes once. Robot and remote come paired from the factory.
2. **Play with the remote.** Joystick drives the robot; the remote's OLED menu on the robot starts games without any phone.
3. **Open the app and sign in** with the parent's Google account, then connect to ELXIE over Bluetooth.
4. **Play the five games** (Colour Quest, Reflex Dash, Echo Memory, Driving Pro, Inverted Drive), six levels each. Progress saves to the account, so a new device continues where the child stopped.
5. **Unlock Elxie Blocks** after the last level of the last game.
6. **Build a first program** from blocks (when started, move forward 1 s, happy face) and press Run; the robot runs it at once.
7. **Explore**: sensors, remote buttons, lights, sounds, pet moods. Live sensor values and the running block are shown in the editor.
8. **Switch back any time**: starting a game or RC hands the robot back to the games; closing Blocks returns to the idle face after 3 s.

MVP features

Every feature below is a must-have for the pilot; anything not listed is out of scope.

Part	Must have in the MVP
Robot hardware	ESP32-S3 (16 MB flash), 2 DC motors (TB6612FNG), 128x64 OLED face, 30 NeoPixel LEDs, buzzer, MPU6050 tilt/motion, QMC5883P compass, HC-SR04 distance, TTP223 touch, 16 IR sensors (line, table edge, corner obstacle: used in Blocks, off in games because of sunlight interference), battery sense
Remote	ESP8266 D1 Mini, joystick + front, bottom and stick buttons, vibration motor, battery sense (mux channel 4), ESP-NOW link to its paired robot
Robot firmware (Home)	One build with games, RC, local OLED menu and the Blocks engine; Bluetooth link to the app (game channel + Blocks channel); remote input for games and blocks; pairing stored in flash; e-stop (remote A+B or 2 s touch)
Web app	Google sign-in, Bluetooth connect, the five games with six levels each, progress saved to the account, Blocks locked until all games are done
Elxie Blocks	Block editor inside the app: motion, face, lights, sound, sensors, remote, pet, control, operators, variables, lists, my blocks; upload, run, stop, live sensors, block highlight while running; 8 program slots on the robot
Pairing	Each robot paired with one remote; at the factory (USB) or at home (touch at power-on + remote A+B at power-on)
Factory tools	Robot and remote testbenches with live web dashboards, one-click flash, factory pairing, robot check

Out of scope

These are deliberately left for after the pilot:

- School edition (Blocks only, class management, teacher view)
- Robot-side unlock code (Blocks is unlocked by the app only during the pilot)
- Wheel encoders and distance-accurate driving; IR sensors in game mode until the sunlight issue is solved
- Saving and sharing block projects in the cloud
- Over-the-air firmware updates (robots are flashed over USB)
- Phone and tablet apps beyond the web app; offline use
- Languages other than English

Status today

The hardware, firmware, Blocks and factory tools are built and proven on robots; the IR sensors are the main open hardware issue.

Part	Status	Evidence / what is left
Robot hardware	Working	Testbench passes IMU, compass, distance, touch, motors; battery reading fixed; IR sensors off in game mode, still available in Blocks (sunlight interference, still unsolved)
Home firmware	Built, boots	A robot that hung at power-on boots after the no-PSRAM fix (Oct 15); still to flash and check all 10; games, RC and pairing confirmed
Blocks on the robot	Working	Tested on the robot: programs upload and run successfully
Remote firmware	Working	Pairing, joystick, buttons, vibration confirmed with the remote testbench
Pairing	Working	Factory pairing OK (robot 90:70:69:16:98:8C with remote E0:98:06:AF:C8:D9)
Web app: games	Working (software team)	Five games, progress in MongoDB, Google sign-in via Clerk
Web app: Blocks	Built, locked	Embedded editor shares the Bluetooth link; unlock-for-testing branch with the software team
Stand-alone Blocks editor	Working	Hosted on GitHub Pages for testing without the app
Factory tools	Working	Robot and remote testbenches with live dashboards, flash, pairing and robot-check scripts

Pilot plan

Proposed: the pilot runs the 10 robots in 10 homes for about four weeks, after a one-week internal shake-down (dates and families to be set).

1. **Prepare (week 0).** Flash all 10 robots with the Home firmware and all 10 remotes with the remote firmware. Run both testbenches on every unit, pair each set at the factory, label robot and remote with the same number, and run a 10-minute smoke test (drive, one game, one Blocks program).
2. **Shake-down (week 1).** The team uses 2 sets at home: full game run, Blocks unlock, 20 programs. Fix blockers, re-flash.
3. **Hand-over (week 2).** One family per set; a 15-minute first setup with the parent; a printed quick-start card.
4. **Run (weeks 2 to 5).** Weekly check-in call; game progress from the app; a short feedback form for the parent and child.
5. **Wrap-up (week 6).** Collect the robots, read logs, decide what goes into the first production batch.

Success criteria

The MVP is done when all of these hold on the 10 pilot sets:

- [ ] All 10 robots and remotes pass both testbenches and are paired 1:1
- [ ] Every robot boots to its face within 5 s on battery, every time, for the whole pilot
- [ ] A parent completes the first setup (sign-in, Bluetooth connect, first game) in under 15 minutes
- [ ] Children finish all five games, and Blocks unlocks for them without help
- [ ] Each child runs at least 5 of their own block programs on the robot
- [ ] Switching between games, RC and Blocks never needs a robot restart
- [ ] No safety issue: e-stop always stops the motors; no injuries or damage reported (robots used on the floor)
- [ ] Feedback form: most children want to keep playing after the pilot

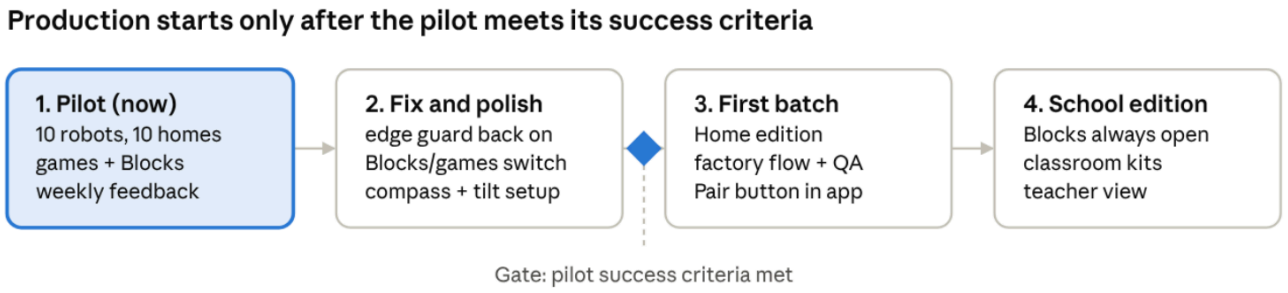
Risks and open issues

The biggest risk is safety: the IR sensors are off in game mode, so games have no table-edge detection, and in Blocks the IR readings can be wrong in sunlight.

Risk or open issue	Impact	Next step
IR sensors off in games, on in Blocks (sunlight interference, still unsolved)	No table-edge guard in games; in sunlight, line, edge, obstacle and "picked up" blocks can misread	Play on the floor, away from direct sunlight (quick-start card); look at modulated IR or shielding
Unit-to-unit hardware differences (one robot hung at boot until PSRAM was disabled)	A robot fails in a family's home	Flash and check all 10 with CHECK_ROBOT.bat before hand-over
Compass needs a one-time spin calibration per robot; pitch and roll may be swapped	Heading and tilt blocks wrong	Add the calibration to factory steps; confirm tilt directions
Blocks unlock is app-only	A child can skip the games via the stand-alone editor	Acceptable for the pilot; robot-side unlock code after the MVP (decide)
Only one Bluetooth connection at a time	Two tabs or devices fight for the robot	Quick-start card: one device per robot

After the MVP

The first production batch waits for the pilot's success criteria; phase lengths are not set yet.



After the MVP · 4 phases, 1 gate

Open decision for phase 3: add the robot-side unlock code, so Blocks cannot be opened without finishing the games.